

Muhammad Haris



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Pakistan

Plant Biotechnologist and Bioinformatician with expertise in transcriptomic and microbiome data analysis, computational plant-microbe interactions, and multi-omics integration using Linux- and R-based bioinformatics pipelines. Proficient in molecular biology techniques, including DNA/RNA extraction, PCR, and qPCR, with additional experience in biological database development and computational tool development.

EDUCATION

Master's in Bioinformatics

Government College University Faisalabad (GCUF)

Faisalabad, Pakistan

Sept. 2025 - Current

CGPA: 3.96 / 4

Expected Date of Graduation: August 2027

Research Focus: Exploring Machine Learning Approaches, Transcriptomics, and Advanced Bioinformatics Analyses for Complex Plant-Microbe Interactions

Bachelor's in Biotechnology

Government College University Faisalabad (GCUF)

Faisalabad, Pakistan

Nov. 2021 - July 2025

CGPA: 3.73 / 4

Thesis: Identification of Therapeutic Targets for Hypertrophic Cardiomyopathy (HCM) Using RNA-seq Data Analysis

- Identified 6 previously unreported therapeutic targets using R and LINUX-based pipelines.

RESEARCH EXPERIENCE

Research Assistant

Functional Genomics & Tissue Culture Lab, GCUF

Faisalabad, Pakistan

Sept. 2024 - Current

- Working on Single-cell RNA sequencing analysis to decipher gene expression in individual cells of a particular plant disease.
- Specialized in maize cell line generation via micropropagation, reproducing plantlets from a single callus for food safety research.
- Provided hands-on training to 4 undergraduate students in tissue culture and molecular biology.
- Worked on 8 manuscripts spanning NGS data analysis, tissue culture, database development, and plant biotechnology under climate change.

Summer Research Intern

International Center for Chemical and Biological Sciences (ICCBS),
University of Karachi

Karachi, Pakistan

June 2025 - August 2025

- Performed DNA library preparation and WES data analysis for Acute Myeloid Leukemia, identifying 884 novel high-impact variants using a custom Linux- and R-based bioinformatics pipeline.

Research Intern

Ayub Agricultural Research Institute (AARI)

Faisalabad, Pakistan

July 2024 - August 2024

- Learned molecular biology (DNA extraction, PCR, PAGE, Microscopy), microbial isolation and tissue culture techniques.

Research Intern

Functional Genomics & Tissue Culture Lab, GCUF

Faisalabad, Pakistan

Jan. 2023 - June 2024

- Trained in advanced laboratory techniques, including DNA extraction, PCR/qPCR, PAGE, microscopy, plant tissue culture, microbial isolation, and biochemical enzyme assays.

SKILLS

Biotechnology: DNA Extraction, qPCR, PCR, Gel Electrophoresis, Bacterial Isolation, Microbiological Techniques, Spectrophotometry, Tissue Culture, Biochemical Enzyme Analysis, Physiological Plant Health Analysis, Pot Experiment, Plant Extract Preparation

Bioinformatics: scRNA-seq Analysis, Variant Calling, WES & NGS Data Analysis, In-silico Primer Designing, In-silico Vector Designing, Phylogenetic Analysis, Web & Database Development, Drug Repurposing, Functional Enrichment, Pathway analysis, Protein Structure Prediction, Protein-protein Interaction

Languages: LINUX, R, Python, SQL, HTML, CSS, and JS

Soft Skills: Strong communication, Teamwork, Leadership, Critical Thinking.

SELECTED PUBLICATIONS

Tissue Culture Optimization: Maize Callus Induction and Factors Affecting the Callus Efficiency

Integrative Plant Biotechnology (2025) | **Published** | [DOI](#)

M. Haris, S. Rasheed, S. Rahman, I. Fatima, A. Mazhar, F. Azeem *

Plant Breeding Under Environmental Uncertainty: A Systematic Review of Challenges in a Changing Climate

New Zealand Journal of Crop and Horticultural Science (2026) | **Published** | [DOI](#)

M. Haris, T. Shaheen*, M. A. Rashid

HCMdrugs: A web-based database cataloging the drugs available for the treatment of Hypertrophic Cardiomyopathy (HCM)

Journal of Computational Biophysics and Chemistry (2026) | **Published** | [DOI](#)

M. Haris, S. Rahman, M. T. Ul Qamar, F. Azeem*

Integrated Physiological and Transcriptomic Analyses Reveal How Biogenic Iron Oxide Nanoparticles Improve Cadmium Tolerance in Wheat Irrigated with Textile Wastewater

Journal of Hazardous Materials (2026) | **Under Review**

M. Haris, F. Batool, S. Hussain, I. Fatima, M.A. Nawaz, A. Mazhar, F. Azeem*

Twenty-Five Years of Genome-Wide Gene Family Analysis: Trends, Tools, and Methodological Shifts - A Meta-Analysis

Genome (2026) | **Under Review**

M. Haris, F. Azeem, K.S. Golokhvast, M. A. Nawaz*

HONORS & ACHIEVEMENTS

- **1st Position in Poster Competition** at Two-day Symposium on Recent Advances in Plant Biotechnology for Sustainable Development & Climate Change, GCUF | May 2026
- **Featured researcher** in ORIC GCUF's inaugural newsletter - Only student selected for this recognition from the entire university | May 2026
- **Prime Minister's** Letter of Appreciation for Environmental Activism
- **Recognition** at the International Youth Conference 9 (IYC 9), UCLA Luskin (USA)
- **Delegate** - Local Conference of Youth (LCOY) 2024
- **Organized** 25+ events and activities on Environmental Awareness and Conservation as the **President** of the Green Youth Movement (GYM) Club, GCUF

DEVELOPED TOOLS, DATABASES & RESOURCES

- **AutoRNAseq** - An Automated end-to-end RNA-seq pipeline integrating differential expression, functional & pathway analysis, PPI networks, GSEA, and ML-based biomarker discovery
- **PGF-Explorer** - Interactive web tool for plant gene family exploration and analysis [!\[\]\(6302aad5aed157b291fddf37b4870784_img.jpg\)](#)
- **HCMdrugs** - Pakistan's first web-based drug database for Hypertrophic Cardiomyopathy [!\[\]\(a9ca2c237943a6d0a9f22252f295b6f3_img.jpg\)](#)
- **Genome-Wide PDF Extractor** - Python-based tool for automated structured data extraction from genome-wide gene family study PDFs [!\[\]\(9a01a64e0b4ff865df7d32ee7991fe8b_img.jpg\)](#)

TALKS & PRESENTATIONS

- **Oral Presentation** on “Integrating Omics Data and Machine Learning for Targeted Gene Discovery in **Maize** (*Zea mays* L.) Biofortification” at 1st International Student Research Conference on "Recent Advances in Bioinformatics and Biotechnology ", GCUF, Faisalabad | **1-2nd July**
- **Oral Presentation** on “AutoRNAseq: An Automated RNA-Seq Data Analysis Tool for Differential Expression and Machine Learning-Based Biomarker Discovery” at 3rd UMT International Conference on Life Sciences (UMTCLS-26), Lahore | **4-6th June, 2026**
- **Poster Presentation** on “Optimization of 2,4-D, Kinetin, L-Proline, and Casein Hydrolysate Concentrations for Embryogenic Callus Induction, Cell Suspension Culture, and Plantlet Regeneration in **Maize** (*Zea mays* L.)” at Two-day Symposium on Recent Advances in Plant Biotechnology for Sustainable Development & Climate Change, GCUF, Faisalabad | **13-14th May, 2026**
- **Poster Presentation** on “Sustainable Reuse of Cadmium-Contaminated Textile Wastewater in **Wheat** Cultivation: A Nano-Physiological and Transcriptomic Perspective” at BSP Biotechnology Congress - 2025 & “National Dialogue on Biotechnology” at the University of Agriculture, Faisalabad | **16-18th Dec 2025**
- **Exhibition** of the *HCMdrugs* database at the GCUF Career Fair 2025, Faisalabad | **15th Oct 2025**
- **Talk** on “Importance of Biodiversity for One Earth” at One-day Seminar on Biodiversity for One Earth: Connecting Civilization through Food, Health, and Energy Resources under Fast Climatic Challenges in Sino-Pak Region, GCUF, Faisalabad | **22nd May 2025**
- **Exhibition** of the *HCMdrugs* database and its future concept at the GCUF Innovation Expo 2025, Faisalabad | **23rd- 24th Jan 2025**

CONFERENCES & WORKSHOPS

5th International Conference On Global Environmental Challenges, Impacts, & Solutions For Sustainability Government College University Faisalabad	01/04/2026 – 02/04/2026
Two-day International Symposium and Hands-on Training Workshop “From Data to Discovery: Mastering NGS and GWAS” Government College University Faisalabad	17/02/2025 – 18/02/2025
6th Hands-On National Training Workshop on Advanced Bioinformatics and Recombinant DNA Technology University of Central Punjab, Lahore	21/08/2024 – 22/08/2024
International Conference on Current Trends, Prospects, and Opportunities in Vaccine Research Centre of Excellence in Molecular Biology, Lahore	28/02/2024–29/02/2024
6th International Conference on Applied Zoology (ICAZ-2023) Government College University Faisalabad	18/12/2023 – 19/12/2023